

Hepatorenal Syndrome

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DEFINITION

Hepatorenal syndrome (HRS) is a potentially reversible functional kidney failure that occurs in patients with acute or chronic liver disease, advanced hepatic failure, and portal hypertension. Although it may occur in patients with subacute liver failure or severe acute alcoholic hepatitis, HRS is mainly observed in patients with advanced cirrhosis. HRS is characterized by impaired kidney function and marked abnormalities in the arterial circulation and endogenous vasoactive systems. In the kidney, there is pronounced vasoconstriction resulting in low glomerular filtration rate (GFR). In the splanchnic circulation, there is marked arteriolar vasodilation resulting in reduction of systemic vascular resistance and arterial hypotension.¹⁻³ Along with cardiocirculatory dysfunction secondary to portal hypertension, systemic inflammation also plays a major role in the pathogenesis of HRS.⁴ Two forms of HRS can be identified on the basis of the progression of the disease (Box 76.1). The acute form (type 1) is characterized by an acute and rapid deterioration in GFR that occurs in the setting of intense systemic inflammation and multiorgan failure (acute-on-chronic liver failure [ACLF]), whereas the chronic form (type 2) has an insidious onset and is characterized by less severe systemic inflammation and moderate kidney failure that follows a steady or slowly progressive course.¹⁻⁴

PSEUDOHEPATORENAL SYNDROME

Pseudohepatorenal syndrome describes concurrent hepatic and kidney dysfunction secondary to a wide variety of infectious, systemic, circulatory, genetic, and other diseases or after exposure to a variety of drugs and toxins (Table 76.1).⁵ These entities must be excluded before the diagnosis of HRS can be established. In these conditions, the liver does not play a causal role in the pathogenesis of kidney failure. Pseudohepatorenal syndrome is usually easy to exclude because the causal agent is frequently known and both kidney and liver functional abnormalities are often found at the initial presentation, without evidence of advanced liver failure or portal hypertension. In contrast, HRS invariably occurs after liver failure and portal hypertension are fully established and frequently develops when the patient is undergoing treatment for these conditions or their complications.

PATHOPHYSIOLOGY AND PATHOGENESIS

Circulatory Dysfunction: Kidney and Systemic Hemodynamic Changes

In HRS, reduction in GFR occurs mainly because of kidney cortical hypoperfusion after intense cortical renal vasoconstriction, which can be demonstrated angiographically as marked beading and tortuosity of the interlobular and proximal arcuate arteries and the absence of a

distinct cortical nephrogram and vascular filling of the cortical vessels (Fig. 76.1). Intense renal vasoconstriction is the final consequence of marked systemic circulatory dysfunction, characterized by progressive splanchnic arterial vasodilation, reduction of the effective arterial blood volume, hypotension, and homeostatic activation of the vasoconstrictor systems.³ This further compromises kidney perfusion because intense vasoconstriction results in blunting of the autoregulation of blood flow, so that kidney perfusion becomes more pressure dependent. In HRS, filtration fraction is also reduced, reflecting a dominant increase in afferent arteriolar tone and a decrease in the ultrafiltration coefficient. Serial systemic hemodynamic studies showed that HRS occurs in the setting of reduced mean arterial pressure (MAP), cardiac output, and wedge pulmonary pressure without change in systemic vascular resistance. These findings suggest that an inability to increase cardiac output to compensate for a decrease in preload (secondary to the accentuation of splanchnic arterial vasodilation) also contributes to the pathogenesis of HRS.⁶ Vasoconstriction is not confined to the renal vascular bed. In HRS it is also observed in other extrasplanchnic territories, including the liver, brain, muscle, and skin.^{2,3}

Neurohumoral Abnormalities

The renal and systemic hemodynamic changes that characterize HRS are a direct consequence of neurohumoral disturbances.^{2,3} Activation of the vasoconstrictor systems (the renin-angiotensin-aldosterone system [RAAS], the sympathetic nervous system [SNS], and vasopressin) is the cause of kidney vasoconstriction; meanwhile, activation of the vasodilator systems occurs mainly in the splanchnic circulation and leads to splanchnic vasodilation. Increases in the serum and urinary levels of vasoconstrictors and in the plasma level of vasodilators are observed in patients with HRS. The vasoconstrictors include renin, norepinephrine, neuropeptide Y, arginine vasopressin, endothelin and F₂ isoprostanes, and urinary cysteinyl leukotrienes; the vasodilators include plasma endotoxin, nitrite (end-product of nitric oxide [NO] metabolism), and glucagon. The sympathetic discharges through the renal nerves are also markedly increased.

In contrast to increased plasma and urinary levels of vasoconstrictors and plasma level of vasodilators, decreased urinary levels of vasodilators have been observed in HRS. These include prostaglandin E₂, 6-keto-prostaglandin F₁ (a stable metabolite of renal prostacyclin), and kallikrein. Because the urinary level of these vasodilators is normal in compensated cirrhosis and higher than normal in decompensated cirrhosis with ascites and normal kidney function, it is postulated that a reduction in the renal synthesis of vasodilators is the final event that leads to the development of HRS.³

Most of these neurohumoral abnormalities found in HRS are also detected, albeit to a lesser extent, in decompensated cirrhosis (with ascites) with normal kidney function and in compensated cirrhosis (without ascites). These findings support the hypothesis that HRS

most likely represents one end of the spectrum of homeostatic abnormalities that occur in liver failure and portal hypertension.

Systemic Inflammation and Metabolic Derangement

Type 1 HRS is often precipitated by bacterial infections, particularly if these infections trigger a severe inflammatory response.³ Inflammation represents a major pathogenic factor for HRS. There is now evidence that type 1 HRS is part of a complex syndrome (ACLF) characterized by the development of organ failure(s) (kidneys, liver, brain, heart, peripheral circulation, gut, lungs, adrenal glands, defensive mechanisms against infections).^{4,7} The mechanism of ACLF has been related to intense systemic inflammation, activation of cytokines, NO and other mediators, oxidative stress, acute deterioration of systemic circulation, and organ failure. Systemic inflammation in cirrhosis mainly derives from the translocation of viable bacteria and/or pathogen-associated molecular patterns (PAMPs) from the intestinal lumen into the intestinal mucosa and the systemic circulation with or without

overt bacterial infection. “Sterile” inflammation can also be caused by the release of danger-associated molecular patterns (DAMPs) by dying hepatocytes in acute hepatic processes (i.e., alcoholic and viral hepatitis). The subsequent sustained activation of innate host immunity leads to the release of proinflammatory cytokines and oxidative stress that impair cardiovascular function and could also damage the kidney and other organs, impairing their function.⁴ Systemic inflammation is associated with blood metabolite accumulation in ACLF, indicating marked alterations in major metabolic pathways, in particular mitochondrial dysfunction, which is involved in the development of kidney and extrarenal organ failure.⁸

Summary of Pathogenesis

Fig. 76.2 shows the pathogenesis of HRS. Liver failure and portal hypertension through pathologic bacterial translocation and endotoxemia increase systemic inflammation and vascular production of vasodilators, including NO, carbon monoxide, and glucagon in the splanchnic circulation, leading to the initiating event of splanchnic arteriolar vasodilation (the peripheral arterial vasodilation hypothesis). Splanchnic vasodilation leads to a decrease in systemic vascular resistance, but MAP is initially maintained by an increase in cardiac output, resulting in a hyperdynamic circulation. Splanchnic vasodilation also decreases arterial filling and reduces the effective arterial blood volume. The subsequent stimulation of the central volume baroreceptors leads to compensatory activation of the vasoconstrictor systems, in particular the arginine vasopressin system, RAAS, and SNS (including its hormones norepinephrine and neuropeptide Y), which help restore effective arterial blood volume. This restoration is achieved in patients with compensated cirrhosis but not in patients with decompensated cirrhosis, in whom progressive splanchnic arteriolar vasodilation leads to increased splanchnic capillary pressure, resulting in an increase in lymph formation that exceeds reabsorption capacity. In parallel, further contraction of the effective arterial blood volume leads to reduction of systemic MAP and further stimulation of the vasoconstrictor systems, resulting in sodium and water retention. The net result of these combined effects is continuous ascites formation (the forward theory of ascites formation).^{2,3}

BOX 76.1 Main Clinical Characteristics of Type 1 and Type 2 HRS

Type 1 HRS

- Acute and rapid deterioration in glomerular filtration rate
- Occurs in parallel with the failure of other organs or systems (e.g., coagulopathy, hepatic encephalopathy)
- In cirrhosis, is a form of acute-on-chronic liver failure
- Frequently follows a precipitating event, mainly bacterial infection
- Rapidly fatal without treatment: mean survival 2 to 3 weeks

Type 2 HRS

- Moderate stable kidney impairment (average serum creatinine 2 mg/dL [176 μ mol/L])
- Mainly causes refractory ascites
- Mean survival without treatment: 6 months

HRS, Hepatorenal syndrome.

TABLE 76.1 Causes of Pseudohepatorenal Syndrome

Potential Causes	Predominantly Tubulointerstitial Involvement	Predominantly Glomerular Involvement
Infections	Sepsis, leptospirosis, brucellosis, tuberculosis, Epstein-Barr virus, hepatitis A virus	Hepatitis B and C viruses, HIV infection, <i>Schistosoma mansoni</i> , liver abscess
Drugs	Tetracycline, rifampin, sulfonamide, phenytoin, allopurinol, fluroxene, methotrexate (high dose), acetaminophen overdose	
Toxins	Carbon tetrachloride, trichloroethylene, chloroform, elemental phosphorus, arsenic, copper, chromium, barium, amatoxins, ^a raw carp bile toxins ^b	
Systemic diseases	Sarcoidosis, Sjögren syndrome	Systemic lupus erythematosus, vasculitis, cryoglobulinemia, amyloidosis
Circulatory failure	Hypovolemic or cardiogenic shock	
Malignancy	Lymphoma, leukemia	
Congenital and genetic disorders	Polycystic liver and kidney disease, nephronophthisis, congenital hepatic fibrosis	
Miscellaneous	Fatty liver of pregnancy, Reye syndrome	Eclampsia, HELLP syndrome, cirrhotic glomerulopathy

^aAccidental poisoning after ingestion of mushrooms of the *Amanita* genus.

^bAccidental poisoning after ingestion of the raw gallbladder or bile of the grass carp (a common practice in rural East Asia).

HELLP, Hemolysis, elevated liver enzymes, low platelet count; HIV, human immunodeficiency virus.

Data from Salerno F, Gerbes A, Ginès P, et al. Diagnosis, prevention and treatment of hepatorenal syndrome in cirrhosis. *Gut*. 2007;56:1310–1318.

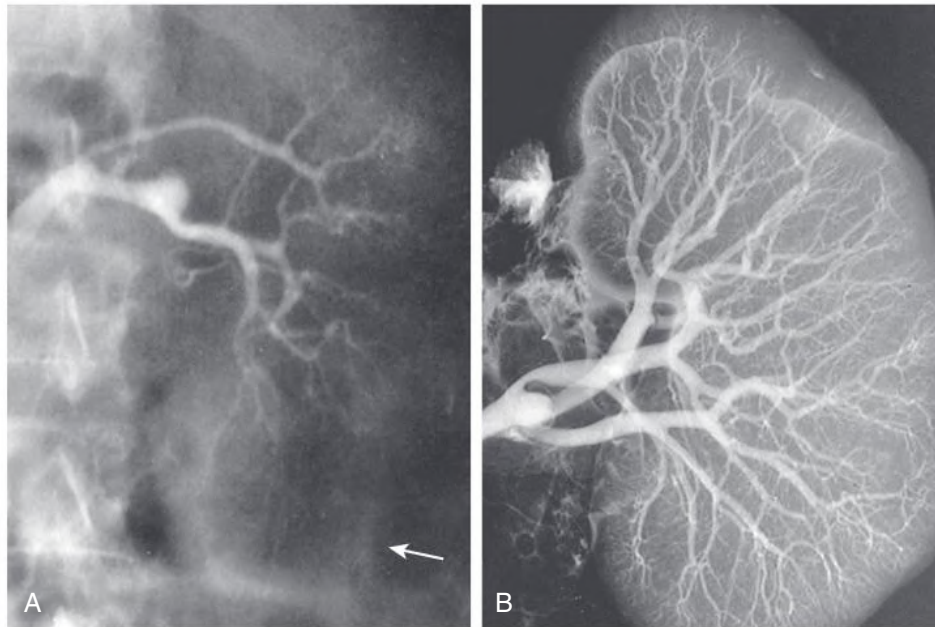


Fig. 76.1 Kidney arteriography of a patient with hepatorenal syndrome (HRS). (A) Kidney angiogram (arrow marks the edge of the kidney). (B) Angiogram performed in the same kidney at autopsy. Note complete filling of the kidney arterial system throughout the vascular bed to the periphery of the cortex. The vascular attenuation and tortuosity seen previously (A) are no longer present. The vessels are also histologically normal; this indicates the functional nature of the vascular abnormality in HRS. (From Levenson D, Korecki KL. Acute renal failure associated with hepatobiliary disease. In: Brenner BM, Lazarus JM, eds. *Acute Renal Failure*. Churchill Livingstone; 1988:535–580.)

The splanchnic circulation is resistant to the effects of vasoconstrictors because of local release of vasodilators; progressive splanchnic vasodilation continues to occur as liver failure and portal hypertension progress. This leads to continued contraction of effective arterial blood volume, which, together with the progressive inability of the cirrhotic heart to respond to reduced preload, results in further reduction of MAP and more intense stimulation of the vasoconstrictor systems. Normally, the effect of vasoconstrictors on the kidney circulation is counterbalanced by the reactive production of intrarenal vasodilators. It is postulated that HRS develops when the balance between vasoconstrictors and vasodilators finally breaks down. The likelihood that this will occur increases with progressive or acute deterioration in liver function or increasing severity of portal hypertension (e.g., after acute alcoholic hepatitis) and is precipitated by events that lead to further volume contraction and reduction of the effective arterial blood volume (e.g., spontaneous bacterial peritonitis [SBP]; see later discussion).

Recently the peripheral arterial vasodilation hypothesis has been revisited. This hypothesis did not take into account that patients with cirrhosis and HRS frequently develop other types of organ failure (e.g., liver, brain, coagulation). This syndrome, known as ACLF, is characterized by systemic inflammation and high short-term mortality.^{4,7} The sustained activation of the innate immune system caused by an abnormal translocation of bacteria and bacterial products from the intestinal lumen (PAMPs) or by DAMPs would lead to the persistent activation of the innate pattern recognition receptors and subsequent chronic inflammation. This inflammatory process would have two major consequences: first, splanchnic arterial vasodilation secondary to local release of endogenous vasodilators, a feature impairing systemic hemodynamics and organ perfusion, and second, the extension of splanchnic inflammation to the peripheral blood and organs, a feature that could damage these organs, contributing to multiple organ failure

(Fig. 76.3).⁴ Therefore, severe systemic inflammation and vasodilation are considered key pathogenic events in the development of HRS in the revisited hypothesis.

EPIDEMIOLOGY

The incidence of HRS in the cirrhotic patient is estimated to be 18% at 1 year and 39% at 5 years.⁹ Neither the cause nor the Child-Pugh score or the model for end-stage liver disease (MELD) score (www.mdcalc.com/child-pugh-score-for-cirrhosis-mortality/) predicts the incidence of HRS. Rather, independent predictors of HRS include dilutional hyponatremia, impaired systemic hemodynamics (high plasma renin activity, noradrenaline concentration, and low cardiac output),^{6,9} abnormal kidney duplex Doppler ultrasound study findings (resistive index > 0.7),¹⁰ and low GFR.³

CLINICAL MANIFESTATIONS

Type 1 and type 2 HRS are considered to be different syndromes rather than different expressions of a common underlying disorder.^{2,3} Type 1 HRS is characterized by a rapid decline in GFR (see Box 76.1) and is observed in patients with acute decompensation of advanced cirrhosis, severe acute alcoholic hepatitis, or subacute liver failure. In addition to developing rapidly progressive acute kidney injury (AKI), patients also develop multiorgan dysfunction, including severe hepatic failure (jaundice, coagulopathy), brain failure (hepatic encephalopathy), and frequently relative adrenal insufficiency. Oliguria is a universal finding, hyponatremia is almost always present, and arterial blood pressure is usually low. Type 1 HRS may be precipitated by bacterial infections (especially spontaneous bacterial peritonitis), severe gastrointestinal (GI) bleeding, or total paracentesis without albumin administration. Left untreated, type 1 HRS tends to rapidly progress, resulting in death of the patient within 2 to 3 weeks.⁹

The Pathogenesis of Hepatorenal Syndrome Type 1

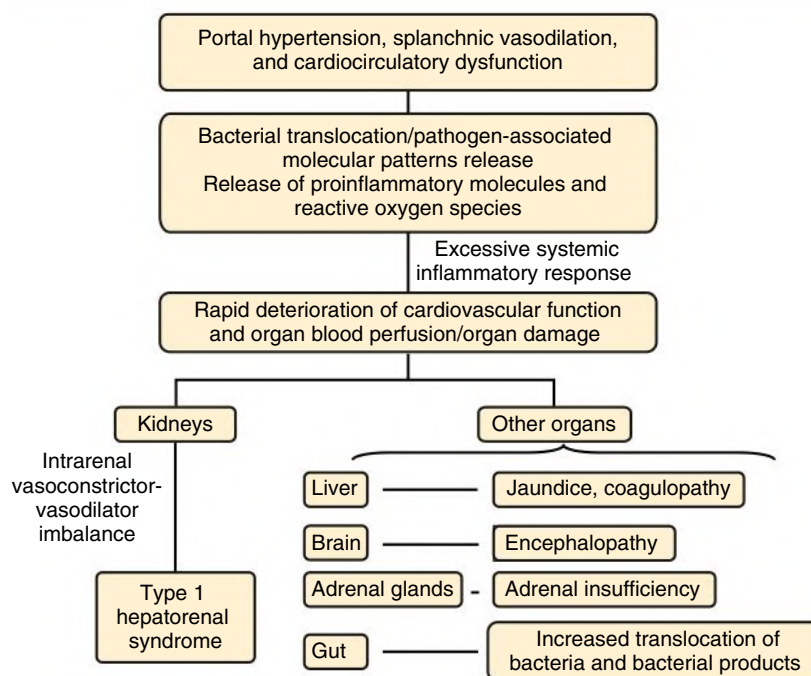


Fig. 76.2 Pathogenesis of Type 1 Hepatorenal Syndrome (HRS). Mechanisms leading to type 1 HRS and multiorgan failure. Patients with cirrhosis and ascites present with severe cardiocirculatory dysfunction that may be further aggravated by infection through the exacerbation of systemic inflammation. Although circulatory dysfunction predominantly affects the kidneys and leads to the development of type 1 HRS, it also decreases the perfusion of other organs and systems such as the liver, with marked impairment in hepatic function and aggravation of portal hypertension; the brain, with the development of hepatic encephalopathy; the adrenal glands, with the development of relative adrenal dysfunction; and the gut, decreasing intestinal motility and promoting intestinal bacterial overgrowth and bacterial translocation. Intense systemic inflammation could also contribute to direct organ damage (immunopathology). (Modified from Arroyo V, Fernandez J, Ginès P. Pathogenesis and treatment of hepatorenal syndrome. *Semin Liver Dis.* 2008;28:81–95; Arroyo V, Fernandez J. Management of hepatorenal syndrome in patients with cirrhosis. *Nat Rev Nephrol.* 2011;7:517–526; and Bernardi M, Moreau R, Angeli P, et al. Mechanisms of decompensation and organ failure in cirrhosis: from peripheral arterial vasodilation to systemic inflammation hypothesis. *J Hepatology.* 2015;63:1272–1284.)

Type 2 HRS is characterized by insidious onset and slowly progressive deterioration of GFR. This is most often observed in patients with cirrhosis and portal hypertension. These patients tend to be less severely jaundiced and mainly present with refractory ascites caused by poor response to diuretics. Low-normal arterial blood pressure, modest prolongation of prothrombin time, and moderate or marked hypoalbuminemia and hyponatremia are usually present. Type 2 HRS tends to progress over months, which most likely reflects the natural course of the disease because additional precipitating factors are not usually identified.^{2,3,9} Mean survival time after onset of type 2 HRS is 6 months.

PATHOLOGY

HRS is mainly a functional kidney disorder, and the presence of significant glomerular and/or tubular disease excludes the diagnosis. However, glomerular abnormalities, including mesangial expansion, capillary wall thickening, mesangial and capillary wall electron-dense deposits, and immune deposits of C3 and immunoglobulin A (IgA), IgM, and IgG, are frequently found in cirrhotic patients with normal kidney function and minimal urinary abnormalities. The presence of such glomerular abnormalities in a cirrhotic patient, therefore, does

not exclude the diagnosis of HRS. Protrusion of the proximal tubular epithelium into Bowman's space (glomerulotubular reflux) is not specific for HRS and is found in other conditions associated with profound kidney ischemia and terminal hypotension. Although early autopsy studies demonstrated normal tubular morphology in patients who had died of HRS, detailed light and electron microscopic studies have documented proximal tubular lesions consistent with ischemic injury. However, these lesions do not explain the low GFR in HRS patients.³ Finally, bile salt-related tubular damage can be observed in patients with HRS and severe cholestasis.¹¹

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

HRS is mainly a diagnosis of exclusion and should be suspected in any patient with subacute or chronic liver disease with advanced liver failure and portal hypertension who develops progressive kidney impairment. In patients with preexisting liver failure, portal hypertension, and kidney failure, the use of nephrotoxic agents (e.g., nonsteroidal anti-inflammatory drugs [NSAIDs] and aminoglycosides) must be stopped, and other conditions leading to kidney failure must be excluded by careful history, physical examination, urine examination, and ultrasound study before the diagnosis of HRS can be considered.

Systemic Inflammation Hypothesis of Cirrhosis Decompensation in Hepatorenal Syndrome and Acute-on-Chronic Liver Failure

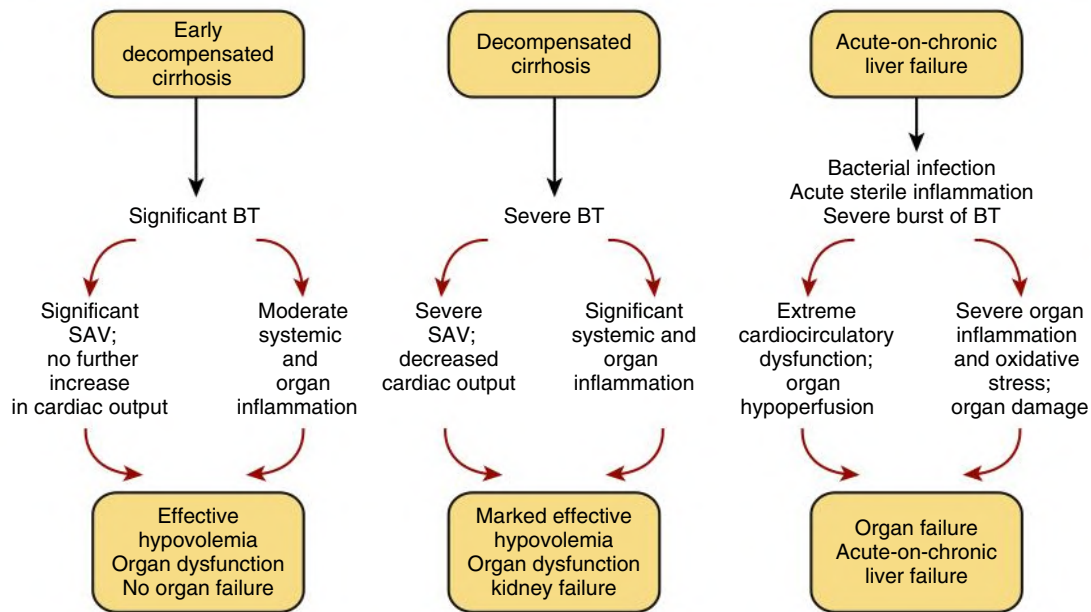


Fig. 76.3 Systemic Inflammation Hypothesis of Cirrhosis Decompensation in Hepatorenal Syndrome and Acute-on-Chronic Liver Failure (ACLF). According to the systemic inflammation hypothesis, bacterial translocation (BT) progressively affects the natural course of cirrhosis, from the compensated stage to hepatorenal syndrome and ACLF. An abrupt increase in systemic inflammation represents the pathophysiologic background for ACLF. SAV, Splanchnic arterial vasodilation. (Modified from Bernardi M, Moreau R, Angeli P, et al. Mechanisms of decompensation and organ failure in cirrhosis: from peripheral arterial vasodilation to systemic inflammation hypothesis. *J Hepatology*. 2015;63:1272–1284.)

The absence of shock or GI bleeding and excess loss of extracellular fluid from the GI tract or urine or from equestration of ascites in the peritoneal cavity must be documented. Prerenal AKI must be excluded by withdrawal of diuretics and fluid challenge with albumin 1 g/kg body weight per day up to the maximum of 100 g/day for 2 days. Proteinuria of less than 500 mg/day and absence of microhematuria help exclude significant coexisting glomerular or tubulointerstitial disease leading to kidney failure and supports the diagnosis of HRS.

The diagnostic criteria for HRS were established by the International Ascites Club in 1996¹ and were revised in 2007, 2015, and 2019.^{11–13} The main revisions were the exclusion of creatinine clearance as a measure of kidney function (because of the difficulty in obtaining accurate urine collection data), the removal of ongoing bacterial infection as an exclusion criterion (so that treatment of HRS is not delayed), the substitution of saline by albumin as the preferred fluid for plasma volume expansion, the removal of the minor criteria and, more importantly, the removal of a fixed cut-off value of serum creatinine (>1.5 mg/dL [$132 \mu\text{mol/L}$], >2.5 mg/dL [$220 \mu\text{mol/L}$] in cases of type 1 HRS) to establish the diagnosis of HRS. This definition of HRS-1 still requires that the diagnosis be established at an advanced stage of AKI (at least stage 2). The most recent revision allows the diagnosis of HRS-AKI in the presence of small increases in serum creatinine (≥ 0.3 mg/dL within 48 hours), therefore promoting the early initiation of vasoconstrictors plus albumin (Box 76.2).¹¹

Serum creatinine and blood urea nitrogen (BUN) are poor markers of GFR in cirrhosis.³ Patients with cirrhosis may have significantly reduced GFR despite normal serum creatinine or BUN because they are frequently malnourished, with reduced lean body mass, and often have a low urea generation rate because of liver failure and low protein intake.

Severe hyperbilirubinemia, which is often present in patients with HRS, interferes with the Jaffe reaction (picric acid) for creatinine quantification and may cause falsely low results. Enzymatic creatinine assays are less susceptible to high bilirubin levels. In cases of uncertainty, GFR may be assessed with use of iodine-125–iothalamate or chromium-51–labeled ethylenediaminetetraacetic acid (^{51}Cr -labeled EDTA). Small studies suggest that serum cystatin C could be an accurate GFR marker in the cirrhotic population,¹⁴ but this requires confirmation.

The most important differential diagnosis for kidney failure in cirrhosis is between type 1 HRS (HRS-AKI) and AKI with acute tubular necrosis (ATN-AKI) because they require rapid therapeutic decisions with different treatments. The parameters traditionally used to differentiate ATN from functional kidney failure (urinary sodium excretion and urinary-plasma osmolality ratio) are classically considered of no value in patients with cirrhosis and ascites.³ However, a fractional excretion fraction of sodium less than 0.2% may help distinguish HRS from ATN.¹¹ Granular casts may be found in the urinary sediment in both HRS and ATN. Recent data suggest that the determination of the urinary levels of neutrophil gelatinase-associated lipocalin (NGAL), a biomarker of tubular damage, could be helpful to differentiate these two entities. More specifically, a cut-off value of $220 \mu\text{g/g}$ of creatinine has been reported to have the best predictive accuracy for ATN diagnosis.^{11,15} Most patients with ATN-AKI have values greater than this threshold, whereas those with HRS-AKI usually have values less than it.¹¹ However, further studies are needed to confirm this finding because this biomarker also can be increased in prerenal acute injury¹⁶ and HRS can progress to ATN. In patients with prolonged HRS-AKI, the kidney is damaged because of chronic ischemia and sustained inflammation. In these cases, granular casts can be observed.

BOX 76.2 New Diagnostic Criteria Hepatorenal Syndrome (HRS-AKI) According to the International Ascites Club

- Cirrhosis, acute liver failure, acute-on-chronic liver failure
- Increase in serum creatinine ≥ 0.3 mg/dL within 48 hours or $\geq 50\%$ from baseline value according to ICA consensus document and/or urinary output ≤ 0.5 mL/kg body weight ≥ 6 hours^a
- No full or partial response, according to the consensus document, after at least 2 days of diuretic withdrawal and volume expansion with albumin. The recommended dose of albumin is 1 g/kg of body weight per day to a maximum of 100 g/day.
- Absence of shock
- No current or recent treatment with nephrotoxic drugs (e.g., nonsteroidal anti-inflammatory drugs, aminoglycosides, iodinated contrast media)
- Absence of parenchymal disease as indicated by proteinuria > 500 mg/day, microhematuria (>50 red blood cells per high-power field), urinary injury biomarkers (if available), and/or abnormal kidney ultrasonography^b
- Suggestion of renal vasoconstriction based on a fractional excretion of sodium of $< 0.2\%$ (with levels $< 0.1\%$ being highly predictive)

^aRequires a urinary catheter.

^bThis criterion is not applicable in cases of known preexisting structural chronic kidney disease (e.g., diabetic or hypertensive nephropathy). From Angeli P, Garcia-Tsao G, Nadim MK, et al. News in pathophysiology, definition and classification of hepatorenal syndrome: a step beyond the International Club of Ascites [ICA] consensus document. *J Hepatology*. 2019;71:811–822.

NATURAL HISTORY

Without treatment, the median survival rate for type 1 HRS is about 2 weeks; that of type 2 HRS is about 4 to 6 months.^{3,9} Patients with type 1 HRS usually die in the setting of multiple organ failure and are now considered to have ACLF, a syndrome recently redefined and characterized by the presence not only of organ failure(s) and high mortality rate but also of systemic inflammation.⁷ Severe hepatic encephalopathy, GI bleeding, and sepsis are common final events in these patients. Patients with type 2 HRS are at higher risk for type 1 HRS after the development of a precipitating event.³

PREVENTION AND TREATMENT

General Principles in the Prevention of Acute Kidney Injury

In patients with cirrhosis and ascites, excessive diuretic therapy results in volume contraction and AKI. To avoid the latter, a stepwise approach to the treatment of ascites is recommended. All patients are advised to follow a low-sodium diet (60–90 mmol/day, equivalent to about 3.5–5 g of salt per day). After this, spironolactone is prescribed at increasing doses (100 mg/day as initial dose; if there is no response within 4 days, 200 mg/day). If required, furosemide is then added at increasing doses every 2 days of 40 to 80 mg/day. Maximum doses of spironolactone and furosemide are 400 and 160 mg/day, respectively. In patients with diuretic resistance, therapeutic paracentesis is indicated but must be combined with plasma volume expansion with use of albumin (8 g/L of ascites removed) to decrease the incidence of circulatory dysfunction after treatment and to prevent development of HRS. The use of renin-angiotensin system blockers and potentially nephrotoxic agents, including NSAIDs, aminoglycosides, and radiocontrast media, should be avoided as much as possible.^{2,3} β -Blockers, used for primary or secondary prophylaxis of variceal bleeding,

reduce MAP and GFR and must be used cautiously in patients with cirrhosis and ascites. Kidney function should be closely monitored, especially in patients with refractory ascites. β -Blockers may increase short-term mortality in this specific cirrhotic subpopulation and should therefore be replaced by band ligation of varices.¹⁷

Preventive Measures

Accepting the hypothesis that HRS represents one end of the spectrum of the homeostatic abnormalities in liver failure and portal hypertension and that it is precipitated by clinical events as volume contraction or bacterial infection, it follows that a major focus of treatment must be to prevent such events and to treat them promptly when they occur. There should be a low threshold for antibiotic therapy for suspected infection in cirrhosis. Spontaneous bacterial peritonitis must be excluded by regular examination of ascites whenever a patient is admitted to the hospital or clinically deteriorates. This infection must be treated not only with broad-spectrum antibiotics but also with albumin infusion (1.5 g/kg at diagnosis and 1 g/kg at day 3) because the latter prevents the subsequent development of HRS and improves short-term survival.¹⁸ A recent randomized controlled trial (RCT) also shows that long-term albumin administration prevents kidney dysfunction and HRS-AKI in patients with cirrhosis and ascites.¹⁹ In contrast, albumin administration is not effective in the prevention of HRS in patients with non-SBP infections nor in hospitalized subjects with acute decompensation and hypoalbuminemia.^{20,21} Primary prophylaxis with norfloxacin has been shown to prevent SBP, delay the development of HRS, and improve survival in cirrhotic patients at high risk for complications (low ascitic protein level < 15 g/L; advanced liver failure with Child-Pugh score ≥ 9 and serum bilirubin ≥ 3 mg/dL or impaired kidney function with serum creatinine ≥ 1.2 mg/dL or serum sodium ≤ 130 mmol/L).²² Norfloxacin is thought to exert its renoprotective effect by reducing the subclinical translocation of viable bacteria and bacterial products from the intestine, systemic inflammation, and NO generation, which, in turn, leads to improved hemodynamics. Prophylactic use of pentoxifylline 400 mg orally three times per day also prevents the development of HRS in patients with acute alcoholic hepatitis, probably by inhibiting the synthesis of tumor necrosis factor- α .²³

General Approach to Treatment

In patients with preexisting liver failure, portal hypertension, and kidney failure, nephrotoxic agents must be stopped. The absence of shock or GI bleeding, as well as fluid loss, must be documented. Prerenal AKI must be excluded by withdrawal of diuretics and fluid challenge with albumin (1 g/kg per day to a maximum of 100g/day for up to 48 hours). Absence of microhematuria and proteinuria also should be confirmed.

Once HRS-AKI has been diagnosed, patients should be assessed for orthotopic liver transplantation (OLT). Suitable candidates should be placed on the waiting list for deceased donor or, if possible, living donor liver transplantation. Bridge treatments are also needed. Pharmacotherapy (see later), transjugular intrahepatic portosystemic shunt, extracorporeal liver support therapy, and (in patients with advanced uremia) kidney replacement therapy should be considered.

Pharmacotherapy

The most promising pharmacotherapy appears to be vasoconstrictors in combination with albumin, targeted at reversal of splanchnic arteriolar vasodilation and restoration of the effective arterial blood volume (Box 76.3). Vasodilators (aiming to reverse renal vasoconstriction) are

BOX 76.3 Pharmacologic Options in the Treatment of Hepatorenal Syndrome

Terlipressin + albumin^a: Terlipressin in IV boluses (0.5–1 mg/4–6 h to start, increasing to a maximum of 2 mg/4 h) or as continuous infusion (starting from 2 mg/24 h to maximum dose of 12 mg/24 h). Treatment is maintained until serum creatinine has reached a final value within 0.3 mg/dL of the patient's baseline serum creatinine. Maximum duration of treatment: 14 days.

Norepinephrine + albumin^a: Norepinephrine IV infusion at 0.5 mg/h to start, increasing the dose by 0.25 to 0.5 mg/h every 4 h up to a maximum of 3 mg/h to achieve an increase in MAP of at least 10 mm Hg. Maximum duration of treatment: 14 days.

Vasopressin + albumin^a: Vasopressin IV infusion at 0.01 U/min to start, increasing the dose upward to a maximum of 0.8 U/min to achieve an increase in MAP of at least 10 mm Hg. Maximum duration of treatment: 11 days.

Midodrine + octreotide + albumin^a: Oral midodrine—2.5 to 7.5 mg/8 h + subcutaneous octreotide 100 µg/8 h to start, increasing midodrine dose to a maximum of 12.5 mg/8 h and octreotide dose to a maximum of 200 µg/8 h to achieve an increase in MAP of at least 15 mm Hg. Maximum duration of treatment: 14 days.

^aAlbumin dose: 1 g/kg on day 1 (up to 100 g) followed by 20 to 40 g/day. Central venous pressure monitoring is advised (but not mandatory) to achieve a value of 10 to 15 mm Hg. IV, Intravenous; MAP, mean arterial pressure.

contraindicated in HRS because they are ineffective and can induce marked hypotension.

Vasopressin analogs exhibit preferential vasoconstrictor action on the splanchnic versus the kidney vascular bed. Terlipressin (triglycyllysine vasopressin) is a synthetic analog of vasopressin that, in addition to having a greater effect on the vascular vasopressin receptors (V_1) than the kidney vasopressin receptors (V_2), as a prodrug requiring transformation to the active form (lysine vasopressin), has a prolonged half-life and can be given as an intravenous (IV) bolus or as continuous IV infusion. Continuous administration improves the tolerability of the treatment (lower incidence of systemic ischemic side effects) and improves its effectiveness (treatment is effective at lower doses).^{3,24} Prospective studies^{25–27} and a large retrospective study²⁸ in patients with type 1 and type 2 HRS have shown that terlipressin combined with daily albumin infusion improved kidney function in 60% of the treated patients, with 37% surviving beyond 1 month (60% of them without OLT). Reversal of HRS was associated with improved survival.^{26,28} The efficacy of terlipressin in the treatment of HRS has been confirmed in five RCTs (Table 76.2).^{29–33} Terlipressin was given at a starting dose of 2 to 6 mg/day, and in three studies the dose was titrated upward on the basis of response to a maximum dose of 12 mg/day. Both treated patients and controls were given daily albumin infusion of 20 to 40 g/day. Reversal of HRS was associated with improved survival in only one study.³¹ The inclusion of patients with more severe kidney failure in the treated group³⁰ and the unexpected high response and survival rates in controls³² may explain why some of these studies did not show a survival benefit of treatment despite success in reversing HRS. The latter is likely a result of the more aggressive use of albumin infusion in these studies because such an approach has been shown to reverse HRS in a high proportion of patients. A meta-analysis confirms that terlipressin plus albumin prolongs short-term survival in patients with type 1 HRS.³⁴ Response to therapy is characterized by a slow and sustained reduction in serum creatinine and improvement in systemic

hemodynamics (marked suppression of the plasma levels of renin and norepinephrine, increase in MAP), urine volume, and serum sodium concentration. Median time to reversal of HRS is 7 days and depends on pretreatment serum creatinine.³

Recurrence of HRS after successful treatment tends to be more common in type 2 (around 50%) than in type 1 HRS (20%).^{25–27,30,32} Retreatment with terlipressin is successful in most patients.^{26,30} Younger age,^{28,31} lower baseline serum creatinine level,³² Child-Pugh score of 12 or lower,^{28,31} administration of albumin,²⁶ serum bilirubin less than 10 mg/dL, and increase in MAP by more than 5 mm Hg after initiation of terlipressin³⁵ are independent predictors of a successful response to treatment. In patients with severe alcoholic hepatitis, the absence of underlying cirrhosis increases the probability of response to treatment. Child-Pugh score (<12)^{26,28,30,31} and MELD score³² are independent predictors for survival. Transient abdominal pain and diarrhea after the first dose of terlipressin treatment are common. Significant ischemic side effects attributed to terlipressin occurred in an average of 4% to 12% of patients.^{25–32} A recent RCT reported a higher rate of severe adverse events, including respiratory failure, in patients receiving terlipressin as IV boluses (10% vs. 3%).³³ Close clinical monitoring of patients (including central venous pressure measurement or noninvasive echocardiographic evaluation of cardiac volume status and function if possible) and administration of terlipressin as continuous infusion are therefore recommended to prevent or rapidly detect the development of any serious adverse event during treatment.

IV vasopressin has been used as an alternative to terlipressin in the treatment of HRS in countries where the latter is not available and was successful in reversing type 1 and type 2 HRS in 42% of patients in a retrospective study.³⁶ Responders had significantly lower mortality and a higher liver transplantation rate than nonresponders. No adverse effect related to therapy was observed.

IV norepinephrine combined with IV albumin and furosemide reversed type 1 HRS in 10 of 12 patients; 3 survived with OLT, and 4 survived for a median of 332 days without OLT.³⁷ With use of a similar dosage regimen, two randomized comparative studies showed that norepinephrine is as effective as terlipressin in the treatment of type 1 and type 2 HRS (see Table 76.2).^{38,39} All recurrences were successfully retreated with the original regimen. Significant treatment-related side effects were low, and the cost of treatment with norepinephrine was 3 and 15 times lower, respectively, than with terlipressin.

Administration of midodrine (an oral vasoconstrictor with α -adrenergic effect), subcutaneous octreotide, and albumin infusion is less effective in improving kidney function in patients with HRS than the combination of terlipressin plus albumin. In a recent RCT, reversal of HRS was achieved in 56% of patients receiving terlipressin plus albumin compared with only 5% in those receiving octreotide plus midodrine and albumin.⁴⁰ If these results are confirmed, octreotide plus midodrine and albumin should be considered the last treatment alternative for patients with type 1 HRS, with terlipressin or norepinephrine combined with IV albumin being the preferred options.

In countries where terlipressin is not available, patients with type 1 HRS should be treated with α -adrenergic drugs (IV norepinephrine or oral midodrine plus subcutaneous octreotide) administered with albumin.

Transjugular Intrahepatic Portosystemic Shunt

Portal hypertension is central in the pathogenesis of the homeostatic abnormalities in HRS. The high operative mortality precludes side-to-side portacaval shunts in HRS patients. The transjugular intrahepatic portosystemic shunt (TIPS) creates a parenchymal tract between branches of the hepatic and portal vein (Fig. 76.4).

TABLE 76.2 Randomized Controlled Trials of Terlipressin or Norepinephrine in Hepatorenal Syndrome

Study	Treatment	Vasoconstrictor Dose	Patients ^a	Reversal of HRS ^b (%)	Survival at 3 Months (%)	Survival at 6 Months (%)
Solanki et al., 2003 ²⁷	Terlipressin + albumin	1 mg/12 h	12 (0)	42 ^c	NA	NA
	Placebo	—	12 (0)	0	0	0
Sanyal et al., 2008 ²⁸	Terlipressin + albumin	1–2 mg/6 h	56 (0)	34 ^b	NA	43
	Placebo	—	56 (0)	13	NA	38
Neri et al., 2008 ²⁹	Terlipressin + albumin	0.5–1 mg/8 h	26 (0)	80 ^b	54 ^b	42
	Albumin	—	26 (0)	19	19	15
Martín-Llahí et al., 2008 ³⁰	Terlipressin + albumin	1–2 mg/4 h	23 (6)	39 ^b	26	NA
	Albumin	—	23 (5)	4	17	NA
Alessandria et al., 2007 ³⁶	Terlipressin + albumin	1–2 mg/4 h	12 (7)	83	67	67
	Norepinephrine + albumin	0.1–0.7 µg/kg/min	10 (6)	70	70	70
Sharma et al., 2008 ³⁷	Terlipressin + albumin	0.5–2 mg/6 h	20 (0)	50	NA	NA
	Norepinephrine + albumin	0.5–3 mg/h	20 (0)	50	NA	NA
Wong et al., 2021 ³¹	Terlipressin + albumin	1–2 mg/6 h	199 (0)	39	49	NA
	Albumin	—	101(0)	18	55	NA

^aNumber of patients with type 2 hepatorenal syndrome in parentheses.

^bSerum creatinine ≤ 1.5 mg/dL.

^c $P < .05$ versus placebo or control.

HRS, Hepatorenal syndrome.

Transjugular Intrahepatic Portosystemic Stent Shunt

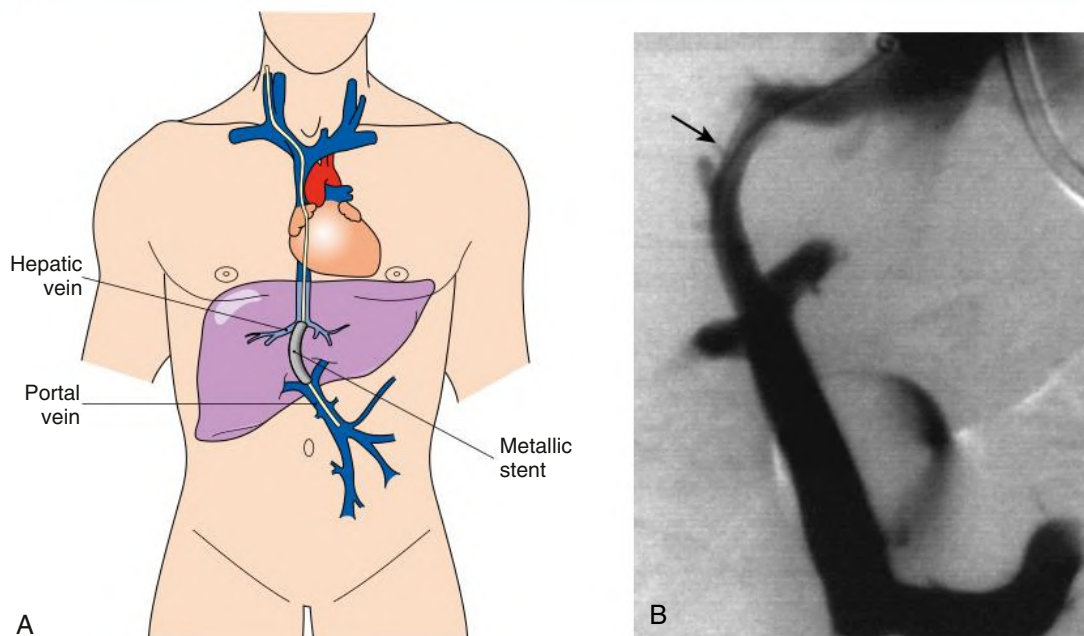


Fig. 76.4 Transjugular Intrahepatic Portosystemic Shunt. (A) An intrahepatic tract has been created between the hepatic vein and the right portal vein. (B) The tract is dilated (arrow) and stented, creating a shunt as demonstrated on shuntogram. (Courtesy Dr. W.K. Tso, Queen Mary Hospital, Hong Kong.)

In experienced hands, operative mortality rates are 1% to 2%, and the morbidity rate is 10%. Procedure-related complications include intra-abdominal bleeding, cardiac arrhythmia, shunt migration and thrombosis, hemolytic anemia, fever, infection, and reactions to radiocontrast media (including nephrotoxicity). The resultant

diversion of portal blood flow from the liver to the systemic circulation may result in transient deterioration of liver function. Shunt stenosis and hepatic encephalopathy are the main long-term complications of TIPS. Exceptionally, TIPS induces disabling encephalopathy. In these cases, closure of the shunt should be performed.⁴¹ In an

earlier study, TIPS improved kidney function in 6 of 7 HRS patients (all with Child-Pugh score <12) and achieved a mean survival of 4.7 (0.3–17) months.⁴² A long-term study of 31 cirrhotic patients with HRS (14 type 1 and 17 type 2) who were not transplant candidates and did not have severe liver failure confirmed that TIPS improved kidney function and survival compared with controls.⁴³ Shunt stenosis and occlusion occurred in 7 patients (treated in 6 by balloon dilation or stent prolongation), and 11 developed hepatic encephalopathy during follow-up. TIPS therefore could be an alternative treatment of type 1 HRS in patients without response to terlipressin/norepinephrine plus albumin.

Combined TIPS and IV terlipressin therapy was performed in nine patients with type 2 HRS.²⁷ All seven patients who responded to terlipressin and relapsed after treatment cessation responded to TIPS. In another study, TIPS was performed in five patients with type 1 HRS after successful treatment with midodrine, octreotide, and albumin.⁴⁴ Complete normalization of kidney function was observed in all patients at 12 months after TIPS. In both studies, improvement or elimination of ascites was an added benefit.

Extracorporeal Liver Support Therapy

Extracorporeal liver support therapy, as a bridge to OLT, relies on biologic (hepatocytes from human or animal source in an ex vivo perfusion system) or nonbiologic methods, including plasma exchange and albumin dialysis systems. Albumin dialysis (molecular adsorbent recirculating system [MARS] or fractionated plasma separation and adsorption [FPSA]) may have beneficial effects in patients with type 1 HRS. In a small RCT, a mean of five treatments with MARS effectively removed albumin-bound toxic metabolites (i.e., bilirubin and bile acids), improved kidney function, and prolonged survival in eight patients (mean survival, 25 ± 5 days) with type 1 HRS and severe liver failure compared with five control patients treated only with hemodiafiltration (mean survival, 5 ± 2 days).⁴⁵ In another study of eight encephalopathic patients with acute alcoholic hepatitis (five with type 1 and two with type 2 HRS), MARS improved kidney function, bilirubin, prothrombin time, grade of encephalopathy, MAP, systemic vascular resistance, and cardiac output, with four patients still alive without OLT at 3 months.⁴⁶ Three large RCTs have evaluated albumin dialysis in patients with ACLF.^{47–49} In the first study, albumin dialysis with MARS was found to be more effective than standard medical therapy in the management of patients with grade 3 or 4 hepatic encephalopathy. Most patients had HRS, and the treatment was found to be safe.⁴⁷ The two other trials compared albumin dialysis with standard medical therapy in patients with type 1 HRS (MARS) or with type 1 and 2 HRS (FPSA). A significant beneficial effect on hepatic encephalopathy was observed in the MARS study but not on survival.⁴⁸ In the FPSA trial, no effect on survival was observed in the whole group or in patients with type 1 HRS, but a significant improvement was observed in patients with high MELD score (>30 points). The administered dosage of dialysis was very low in both studies (six sessions of 6 hours in 21 days). In these studies, therapy was well tolerated.^{48,49} Further studies are needed to ascertain the potential role of albumin dialysis in type 1 HRS. Patients with type 2 HRS are usually treated by total paracentesis or TIPS.

Kidney Replacement Therapy

Hemodialysis and peritoneal dialysis are difficult in HRS patients with advancing uremia. Conventional hemodialysis is hampered by the invariable systemic hypotension, and the efficacy and safety of peritoneal dialysis have not been well evaluated in these patients.

Continuous venovenous hemofiltration (CVVH) or continuous venovenous hemodiafiltration (CVVHDF) has been advocated for the treatment of advancing uremia in HRS,⁵⁰ especially patients with fulminant hepatic failure in whom intermittent dialysis can increase intracranial pressure. Anticoagulation should be minimized or avoided (especially in patients with severe coagulopathy) by giving the replacement fluid in the predilutional mode. When anticoagulation is needed, conventional heparin is generally recommended. Regional citrate anticoagulation is not recommended in advanced cirrhosis because the liver plays a significant role in citrate metabolism. If used, dose adaptation and close metabolic monitoring are required, especially after prolonged use. Close serum calcium monitoring is mandatory because hypocalcemia is common because of impaired citrate clearance. Bicarbonate should be used instead of lactate as the buffer for the replacement solution to minimize metabolic acidosis. MARS may be combined with either CVVH or hemodialysis for the treatment of HRS, especially in patients with severe hepatic encephalopathy.

Liver Transplantation

Liver transplantation is the treatment of choice in patients with advanced cirrhosis, including those with type 1 and type 2 HRS.^{2,3,51,52} The hemodynamic and neurohormonal abnormalities associated with HRS disappear within the first month after liver transplantation. For this reason, calcineurin inhibitors should be withheld in the first few days after OLT to give the ischemic kidneys a chance to recover. Patients with HRS who undergo transplantation have more complications, spend more days in the intensive care unit, and have a higher in-hospital mortality rate than transplantation patients without HRS. The long-term survival of patients with HRS who undergo liver transplantation, however, is good (3-year survival 60%) compared with transplantation in patients without HRS (70%–80%).⁵¹ Although kidney function improves after transplantation in HRS patients, it never reaches that observed in non-HRS patients.⁵³ The incidence of end-stage kidney disease in patients with HRS is slightly higher than that observed in transplantation recipients without HRS (7% vs. 2%). Thus, OLT is associated with comparable liver outcome but inferior kidney outcome in patients with HRS. This problem cannot be overcome by performing combined kidney and liver transplantation in HRS, which overall produces outcomes no better than with OLT alone.⁵⁴ However, in the rare subgroup of patients with HRS who required prolonged pretransplant dialysis for more than 2 months, combined kidney and liver transplantation did confer an advantage in patient survival and use of hospital resources. These infrequent patients usually have normal kidney size and morphology on ultrasound but develop some additional indications of organic nephropathy during follow-up (mild proteinuria or microhematuria).⁵⁵

The main problem with liver transplantation in type 1 HRS is that most patients die before transplantation. The introduction of the MELD score for listing has partially solved the problem because patients with HRS are prioritized for transplantation. As mentioned earlier, treatment of HRS with vasoconstrictors and albumin improves survival in the group of patients with response to treatment, increasing the number of patients reaching liver transplantation. In addition, reversal of HRS before transplantation may decrease early morbidity and mortality after transplantation and prolong survival.⁵²

Therapeutic Algorithm

Fig. 76.5 shows the recommended algorithm for the management of HRS. Recent recommendations have been published by the

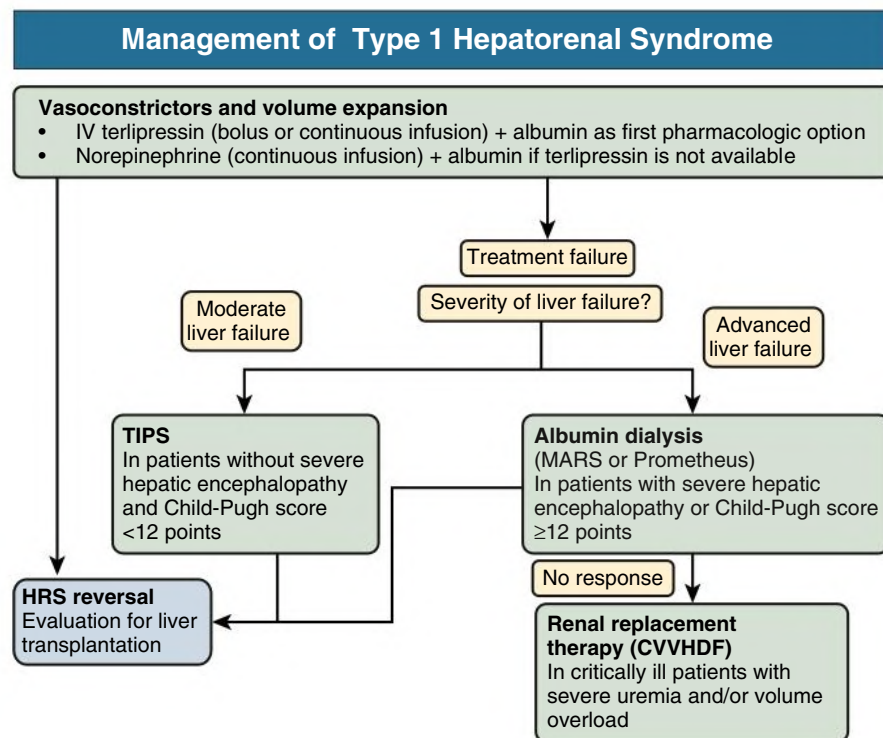


Fig. 76.5 Algorithm for the Management of Type 1 Hepatorenal Syndrome (HRS). Initial treatment is based on the use of vasoconstrictor therapies plus albumin. Intravenous (IV) terlipressin is the most established and is the preferred therapy. If terlipressin is not available, continuous IV norepinephrine should be used. Patients with history of significant atherosclerotic or cardiovascular disease should be treated with norepinephrine. If pharmacologic therapy fails, the severity of liver failure should be assessed. In patients with less severe liver dysfunction (bilirubin < 5 mg/dL or 85.5 μ mol/L and Child-Pugh score < 12) without severe hepatic encephalopathy (grade $\leq$ 2) or history of recurrent encephalopathy, transjugular intrahepatic portosystemic shunt (TIPS) should be considered. In patients with more severe liver failure (Child-Pugh score $\geq$ 12) and/or severe hepatic encephalopathy (grade > 2), albumin dialysis (molecular adsorbent recirculating system [MARS] or fractionated plasma separation and adsorption [FPSA]) should be considered. In critically ill cirrhotic patients with advancing kidney failure, continuous venovenous hemodiafiltration (CVVHDF) is the treatment of choice.

International Ascites Club.^{11,56} OLT is undoubtedly the treatment of choice for patients with HRS, but other treatments described earlier must be used as a bridge to OLT and may improve the kidney outcome after successful OLT. In patients who are not transplantation candidates, these treatments are their only chance for increased survival and in some cases may improve their condition to an extent that may allow them to be reconsidered for transplantation. The choice of therapeutic modalities depends on the availability of resources and expertise on the one hand and the severity of underlying kidney and liver failure and the general condition of the patient on the other. All patients should be considered for vasoconstrictor therapy combined with albumin infusion. Among the vasoconstrictor therapies, IV terlipressin, combined with daily albumin infusion, is most established and is the preferred therapy.^{2,3,56} In countries where terlipressin is not available, continuous IV norepinephrine or vasopressin may be used as an alternative. Patients with a history of significant atherosclerotic or cardiovascular disease should be treated with norepinephrine plus

albumin. Oral midodrine combined with subcutaneous octreotide and albumin is another alternative in the treatment of HRS, especially in patients with type 2 HRS. In patients with relatively well-preserved liver function (serum bilirubin < 5 mg/dL or 85.5 μ mol/L and Child-Pugh score < 12) without severe hepatic encephalopathy (grade $\leq$ 2) or history of recurrent encephalopathy, concurrent severe bacterial infection, or serious cardiovascular or pulmonary disease, TIPS should be considered, especially in patients with recurrence after vasoconstrictor therapy, a situation more commonly observed in HRS type 2 patients.³ In these patients, TIPS may have the added benefit of relieving refractory ascites. TIPS also appears to achieve complete normalization of kidney function in selected patients after an initial successful response to vasoconstrictor therapy. In patients with severe liver failure (Child-Pugh score $\geq$ 12) and severe hepatic encephalopathy (grade > 2), MARS should be considered. In critically ill cirrhotic patients with advancing kidney failure, CVVH is the treatment of choice and may be combined with other therapeutic modalities, especially MARS or FPSA.

SELF-ASSESSMENT QUESTIONS

1. Which of the following pathogenic events is/are usually not observed in patients with HRS?
 - A. Splanchnic vasodilation
 - B. Cardiac dysfunction
 - C. Systemic inflammation and mitochondrial dysfunction
 - D. Tubular and glomerular lesions
2. Diagnosis of HRS is established by which one of the following?
 - A. Liver biopsy
 - B. Kidney biopsy
 - C. Measurement of plasma renin activity
 - D. It is a clinical diagnosis and is established by exclusion.
3. Which affirmation(s) is/are incorrect in patients with hepatorenal syndrome?
 - A. It occurs in patients with subacute or chronic liver failure and portal hypertension.
 - B. Patients are normally hypertensive.
 - C. Plasma renin activity is usually high.
 - D. It can be observed in patients with and without ascites.
 - E. Patients can be infected.
4. Which affirmation(s) is/are incorrect regarding the treatment of patients with HRS?
 - A. The combination of terlipressin and albumin is the preferred treatment.
 - B. TIPS is an alternative treatment.
 - C. Patients must undergo early dialysis.
 - D. Albumin dialysis can be used in patients in whom pharmacologic treatment has failed.
 - E. The combination of midodrine, octreotide, and albumin is a very effective pharmacologic option.
5. Which affirmation(s) is/are correct in the prophylaxis of hepatorenal syndrome?
 - A. Norfloxacin prevents the development of HRS in patients with advanced cirrhosis and low-protein ascites.
 - B. Albumin infusion (1.5 g/kg at diagnosis and 1 g/kg at day 3) prevents the development of HRS and improves short-term survival in patients with SBP.
 - C. Albumin infusion prevents HRS in patients with infections other than SBP.
 - D. Oral midodrine prevents HRS.

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